⚡ AMD Developer Hackathon: ACT III

# DocuSwarm

Parallel Document Due-Diligence on AMD GPUs

Team Yanero Rockstar · Jose Miranda · October 2026

THE PROBLEM

# Reviewing contracts is slow, manual work

- Legal and compliance teams review contracts, NDAs, and reports **one document at a time**.
- Spotting risky clauses — termination penalties, liquidated damages, non-competes — takes hours per batch.
- Most "AI document tools" just call an LLM once per document, sequentially — no faster than a human skimming.
- Nothing shows whether the underlying infrastructure can actually scale.

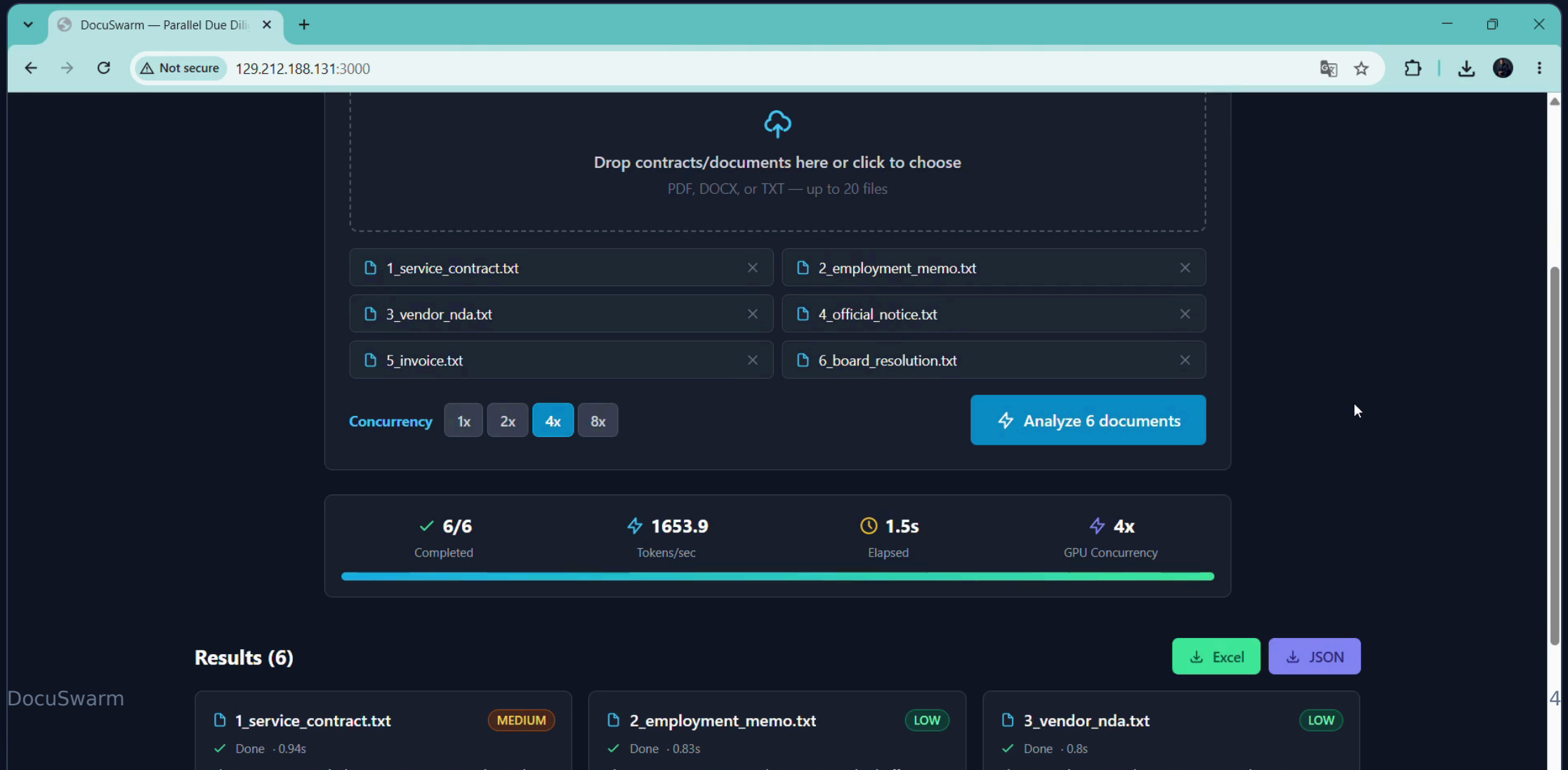
THE SOLUTION

# Upload many documents. A swarm analyzes them in parallel.

- Drop in up to 20 documents — contracts, NDAs, invoices, notices.
- Every document gets its own async agent task, bounded by a concurrency semaphore (1x–8x).
- All agents hit the **same vLLM endpoint on an AMD Instinct GPU** at the same time.
- Live dashboard streams risk level, red flags, key dates, and parties per document — plus real GPU throughput.
- One-click export to Excel / JSON for the compliance team.

LIVE DEMO

# 6 documents, analyzed concurrently, in real time



REAL AMD GPU THROUGHPUT — NOT A SIMULATION

# Measured live on an AMD Instinct MI300X

| 6/6 | 1.5s | 1,653.9 | 4x |
| --- | --- | --- | --- |
| documents completed | total elapsed time | tokens / second | GPU concurrency |

Served by vLLM running **Qwen3-8B** on an **AMD Instinct MI300X** (AMD Developer Cloud) — this is the exact continuous-batching workload AMD Instinct GPUs + ROCm + vLLM are built for.

ARCHITECTURE

# Provider-agnostic by design

**React + Vite Dashboard**
File upload, concurrency selector, live SSE-streamed metrics & results

↓ multipart upload / SSE stream ↓

**FastAPI + asyncio Backend**
One async task per document, bounded by a semaphore (swarm orchestrator)

↓ N concurrent chat requests ↓

**vLLM on AMD Instinct GPU (ROCm)**
Continuous batching serves all concurrent requests on one GPU endpoint

The backend talks to **any OpenAI-compatible endpoint** — moving from a laptop (mock mode) to a live AMD GPU instance is a **one-line environment variable change**, no code changes.

This is what let us go from local development to a real AMD Instinct MI300X deployment without touching the application code.

WHY THIS FITS THE HACKATHON

# Built to make GPU throughput visible, not claimed

- **Real concurrency, not a queue.** Documents don't wait in line — they all hit the GPU together.
- **A live performance dashboard.** Tokens/sec and elapsed time are read from vLLM's real usage field.
- **Genuine LLM judgment.** Risk levels come from the model reasoning about each document, not keyword matching — e.g. a contract flagged MEDIUM risk with specific red flags (termination penalty, liquidated damages).
- **Production-shaped, not a toy demo.** Async orchestration, structured export, and a deployment guide for AMD Developer Cloud.

TECH STACK

# Everything real, nothing mocked in the final demo

FastAPI + asyncio

React + Vite

vLLM

Qwen3-8B

AMD Instinct MI300X

AMD ROCm

AMD Developer Cloud

openpyxl (Excel export)

# Thank you

DocuSwarm — Parallel Document Due-Diligence on AMD GPUs

github.com/yanerox69/yanerodata

Team Yanero Rockstar — AMD Developer Hackathon: ACT III